

Lecture 05

Taxes

Ivan Rudik

AEM 4510

Roadmap

1. What are the different kinds of price instruments in theory and the real world?
2. What happens under a tax?
3. Why cost-effectiveness requires equimarginal abatement when firms have heterogeneous MACs

Taxes

Emission taxes



Emission taxes

The main tax we will focus on is an **emission tax**

When a polluter faces an emission tax, it must pay a fee τ to the regulator for each unit of emissions E

It is as if the government claimed the property rights to the air on behalf of the citizens, and then sold the rights to pollute back to the firm at price τ

Solves the two big problems with externalities:

1. Poorly defined property rights (assigned to the regulator)
2. High transactions costs (pay a flat fee)

Emission taxes

What is the socially optimal tax?

The τ that minimizes total social cost: abatement cost + damages (ignore tax payments since they're just a transfer from firms to the regulator)

Let $C(E)$ be abatement cost of emissions level E and $D(E)$ be damages at E

We know abatement costs are decreasing in emissions and damages are increasing in emissions

Emission taxes

To figure out the socially optimal tax we need to know how the firm and regulator make decisions

First lets set up the firm problem: they want to minimize the cost of satisfying the policy

Emission taxes: firm

The firm's problem is then:

$$\min_E C(E) + \tau E$$

The firm's first-order condition to minimize costs is:

$$\underbrace{-C'(E^*)}_{\text{MAC}} = \tau$$

The firm minimizes costs by choosing emissions E^* so that its MAC equals the tax

Emission standards: firm

The firm minimizes costs by choosing emissions so that its MAC equals the tax

This makes sense!

The tax is the PMC of emitting, the MAC is the PMB of emitting (reduced abatement cost)

Costs are minimized when these two things are equal

Emission taxes: regulator

The regulator's problem is to maximize social welfare / minimize total costs of abatement and damages

We already know this occurs when:

$$-C'(E^*) = D'(E^*)$$

but we know that firms select E^* so that:

$$-C'(E^*) = \tau$$

This means that the regulator's optimal tax is given by:

$$\tau^* = D'(E^*)$$

Emission taxes: regulator

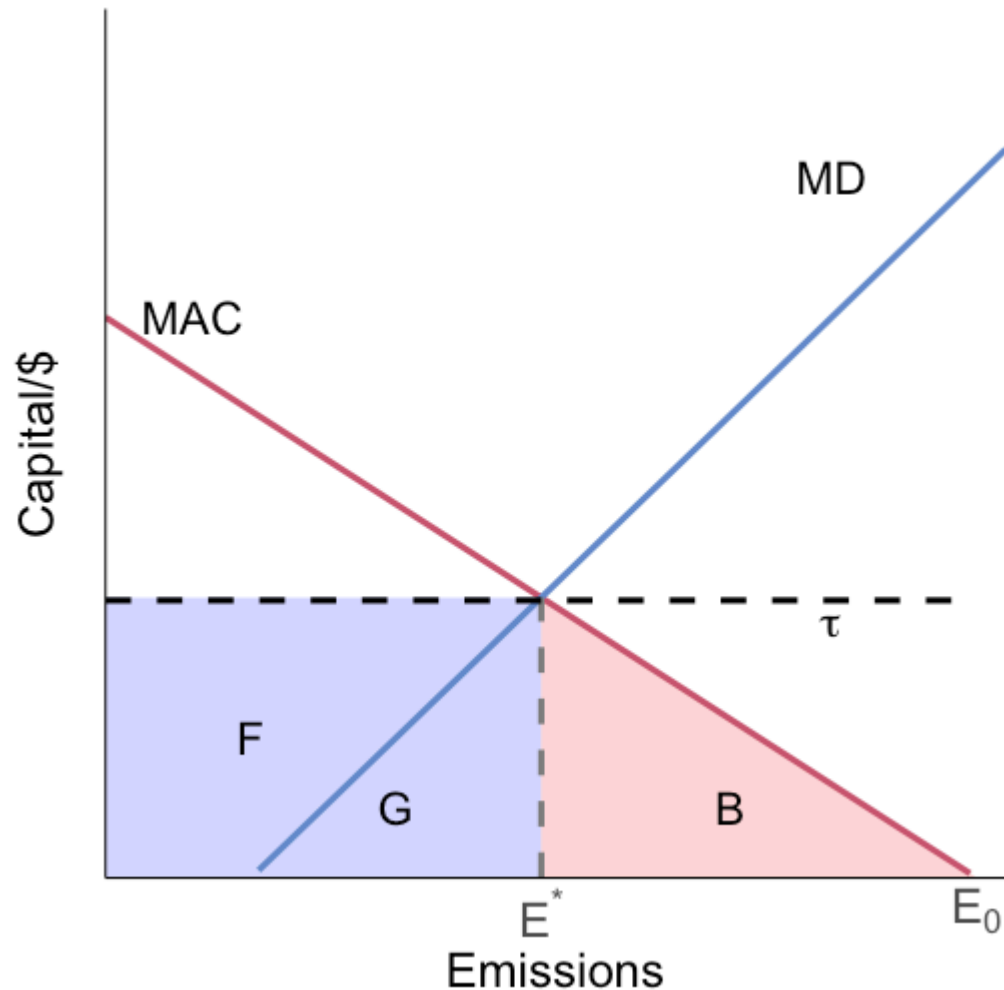
The optimal tax is equal to the MD **at the optimal emission level**

This also makes a lot of sense

We want the firm to act as if they are bearing the damage costs on the margin

The tax actually makes them bear a cost exactly equal to marginal damage

Emission taxes: graphical



The optimal tax equals MD at E^*

You can think of this as the firm being forced to pay for damages equal to the cost of the last unit of emissions

In addition to paying abatement cost equal to the red area B , the firm also has a tax payment equal to the blue area $F + G$

Emission taxes: real world

Australia started carbon tax on July 1, 2012

\$23 AUD per ton of carbon emitted for large emitters as a response to the Copenhagen Accord of 2009

Australia hopes to reduce carbon emissions by 80% below 2000 levels by 2050

Started by Julia Gillard government and revoked by Tony Abbott government in July 2014

Emission taxes: real world

In 2008, the province of British Columbia implemented North America's first broad-based carbon tax

The carbon tax applies to the purchase and use of fossil fuels and covers approximately 70% of provincial greenhouse gas emissions

Beginning April 1, 2018, B.C.'s carbon tax rate is \$35 per tonne of carbon dioxide equivalent emissions

To improve affordability, government increased the Climate Action Tax Credit

Subsidies

Abatement subsidies

What is the socially optimal abatement subsidy s ?

The s that minimizes total social cost: abatement cost + damages (ignore tax payments since they're just a transfer from firms to the regulator)

Let $C(E)$ be abatement cost of emissions level E and $D(E)$ be damages at E

First lets set up the firm problem: they want to minimize the cost of satisfying the policy

Abatement subsidies: firm

The firm's problem is then:

$$\min_E C(E) - s(E_0 - E)$$

where we are substituting in $E_0 - E$ for A

The firm's first-order condition to minimize costs is:

$$-C'(E^*) = s$$

The firm minimizes costs by choosing emissions E^* so that its MAC equals the subsidy rate!

Abatement subsidies: firm

The firm minimizes costs by choosing emissions so that its MAC equals the subsidy rate

This makes sense!

The subsidy is the MC of emitting: the forgone abatement subsidy

The MAC is the marginal benefit of emitting (reduced abatement cost)

Costs are minimized when these two things are equal

Abatement subsidies: firm

The regulator's problem is to minimize the social cost of emissions, we know they want to set:

$$-C'(E^*) = D'(E^*)$$

but we know that firms set:

$$-C'(E^*) = s$$

This means that the regulator wants to set:

$$s^* = D'(E^*)$$

Emission taxes: regulator

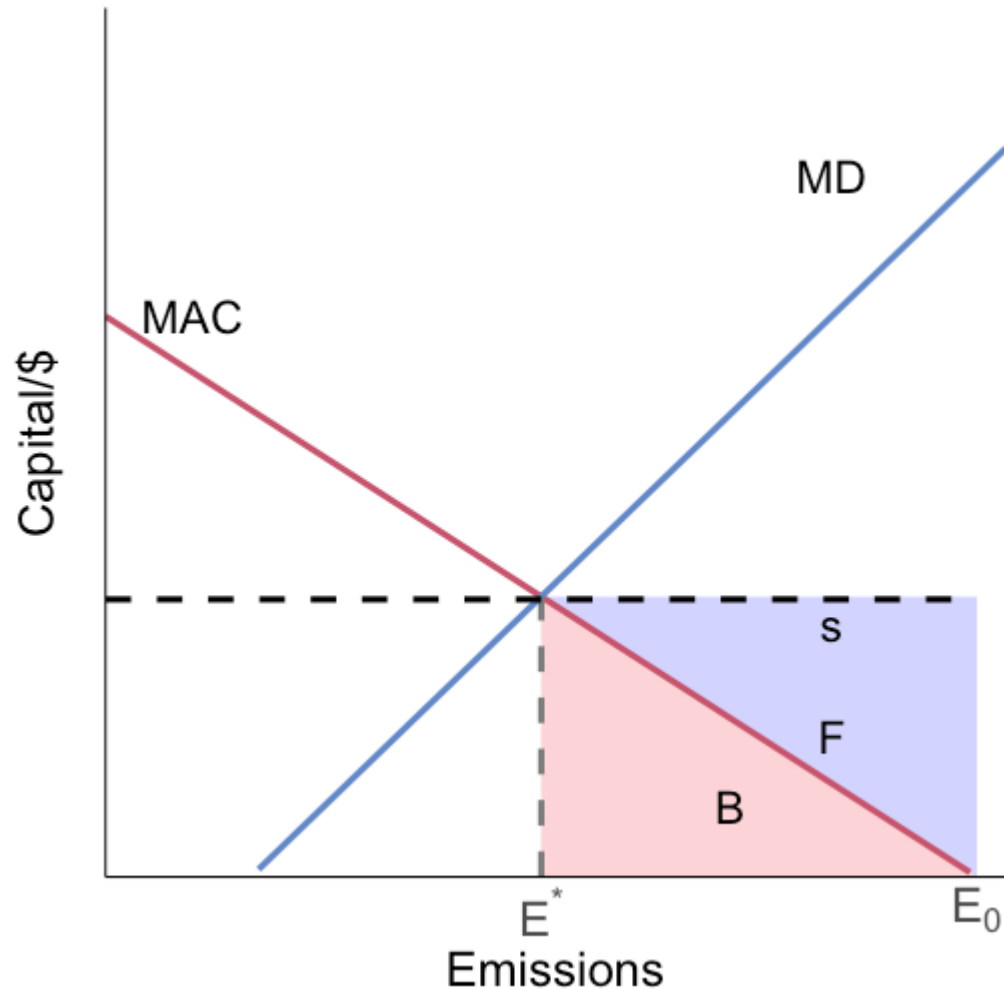
The optimal subsidy is equal to the MD at the optimal emission level

This also makes a lot of sense

We want the firm to act as if they are bearing the damage costs on the marginal

The subsidy actually makes them bear a cost exactly equal to marginal damage

Abatement subsidies: graphical



The optimal subsidy equals MD at E^*

You can think of this as the firm being forced to pay for damages equal to the cost of the last unit of emissions

In addition to paying abatement cost equal to the red area B , the firm also has subsidy **revenues** equal to the red+blue area $B + F$

Abatement subsidies

It is as if the government claimed the property rights to the air on behalf of the firm, and then pays the firm for the right to clean air on behalf of the citizens

Congestion Pricing

Congestion pricing

Congestion pricing is a Pigouvian tax on driving in high-traffic areas

Drivers impose external costs on others:

- Traffic delays
- Air pollution
- Noise
- Accident risk

A toll internalizes these externalities by making drivers pay the social cost of their trips

NYC congestion pricing

In January 2025, NYC became the first US city to implement congestion pricing

How it works:

- \$9 toll for cars entering Manhattan south of 60th Street (peak hours)
- \$2.25 for motorcycles, \$14.40 for small trucks, \$21.60 for large trucks
- Off-peak and overnight discounts
- Exemptions for emergency vehicles, buses, residents below income threshold

NYC congestion pricing

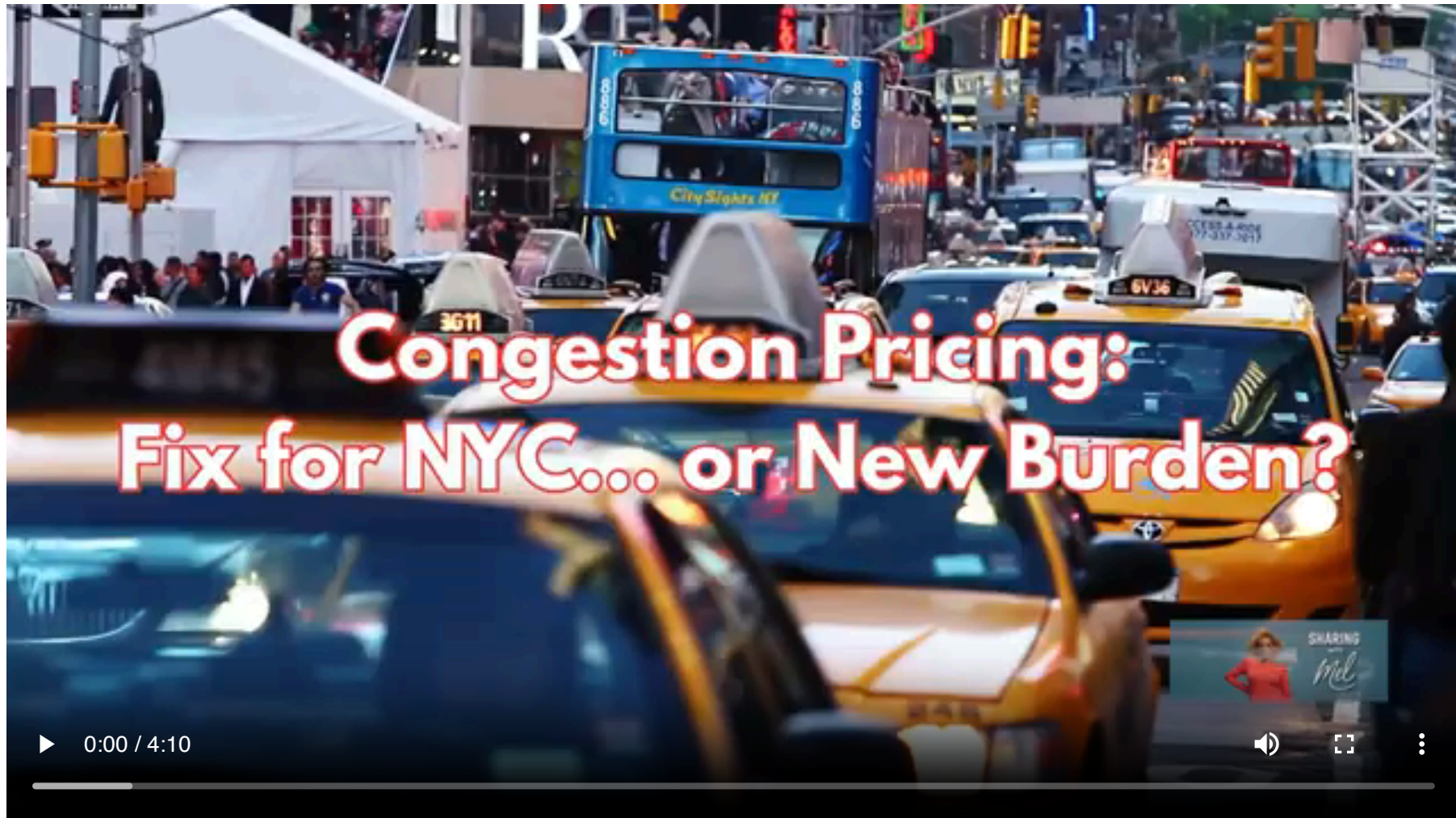
Goals:

- Reduce traffic by 10-15%
- Raise \$1 billion annually for MTA public transit improvements
- Faster bus and emergency vehicle travel times
- Improved air quality in Manhattan

NYC congestion pricing explained

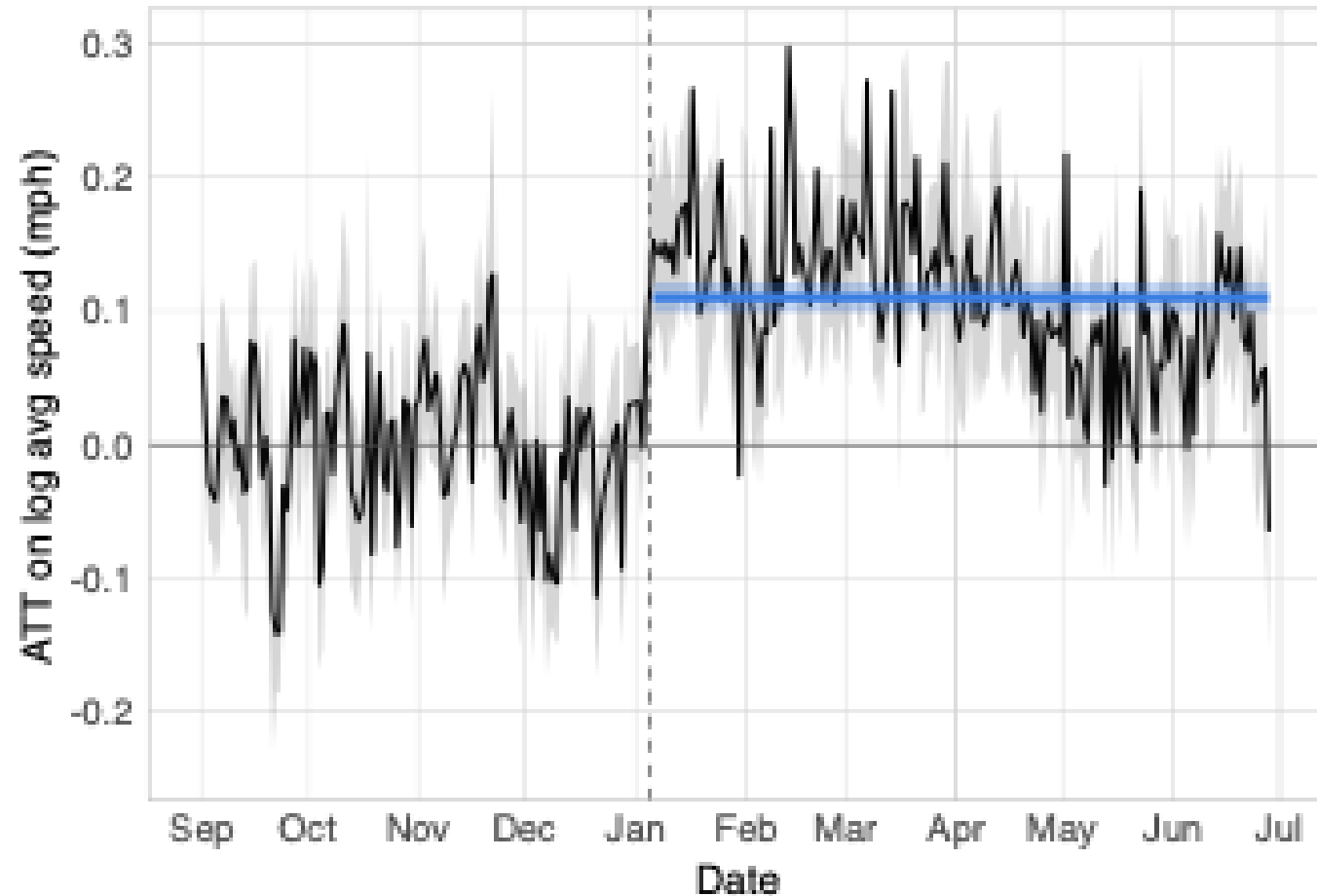


NYC congestion pricing: one year later



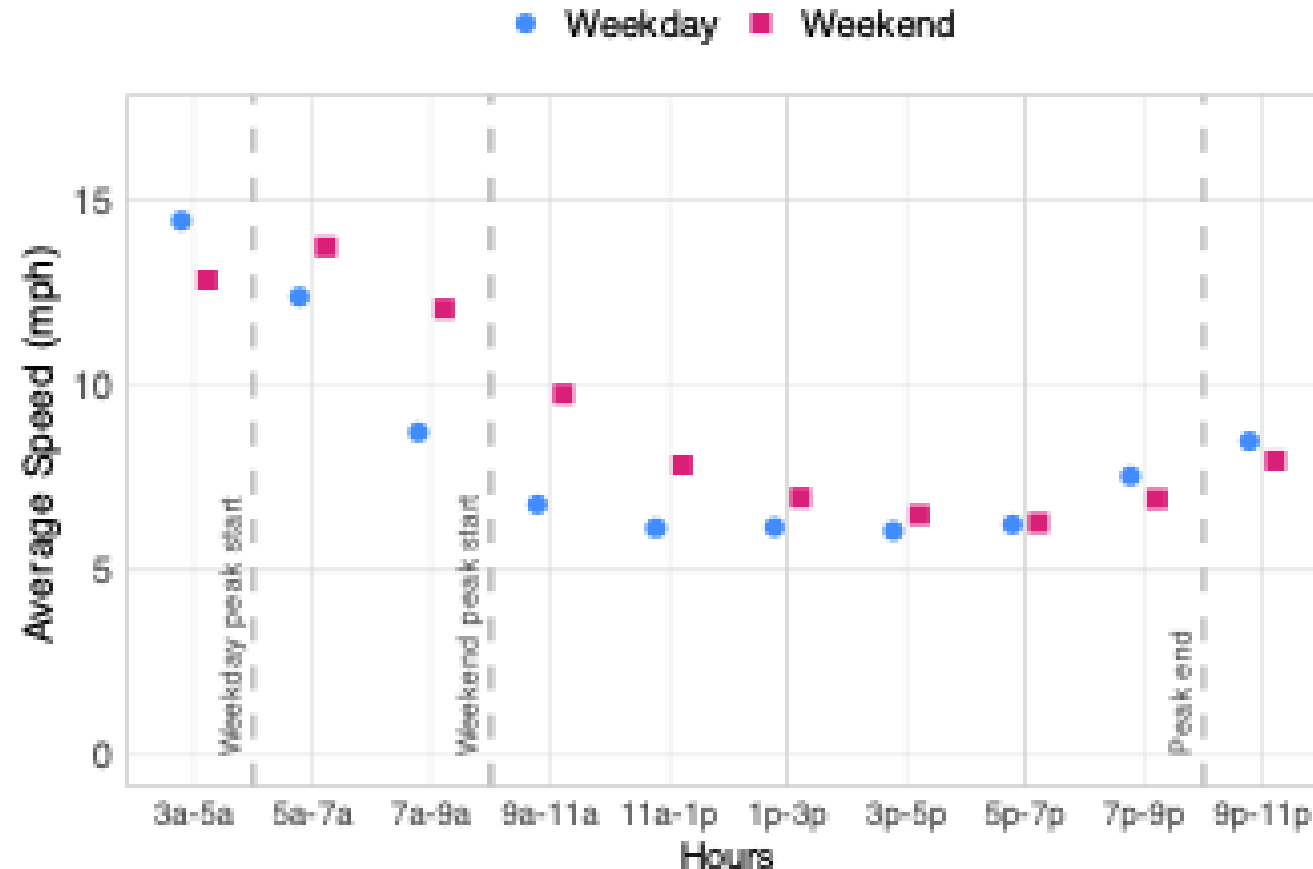
NYC congestion pricing: one year later

(a) ATT on log speeds: road segments



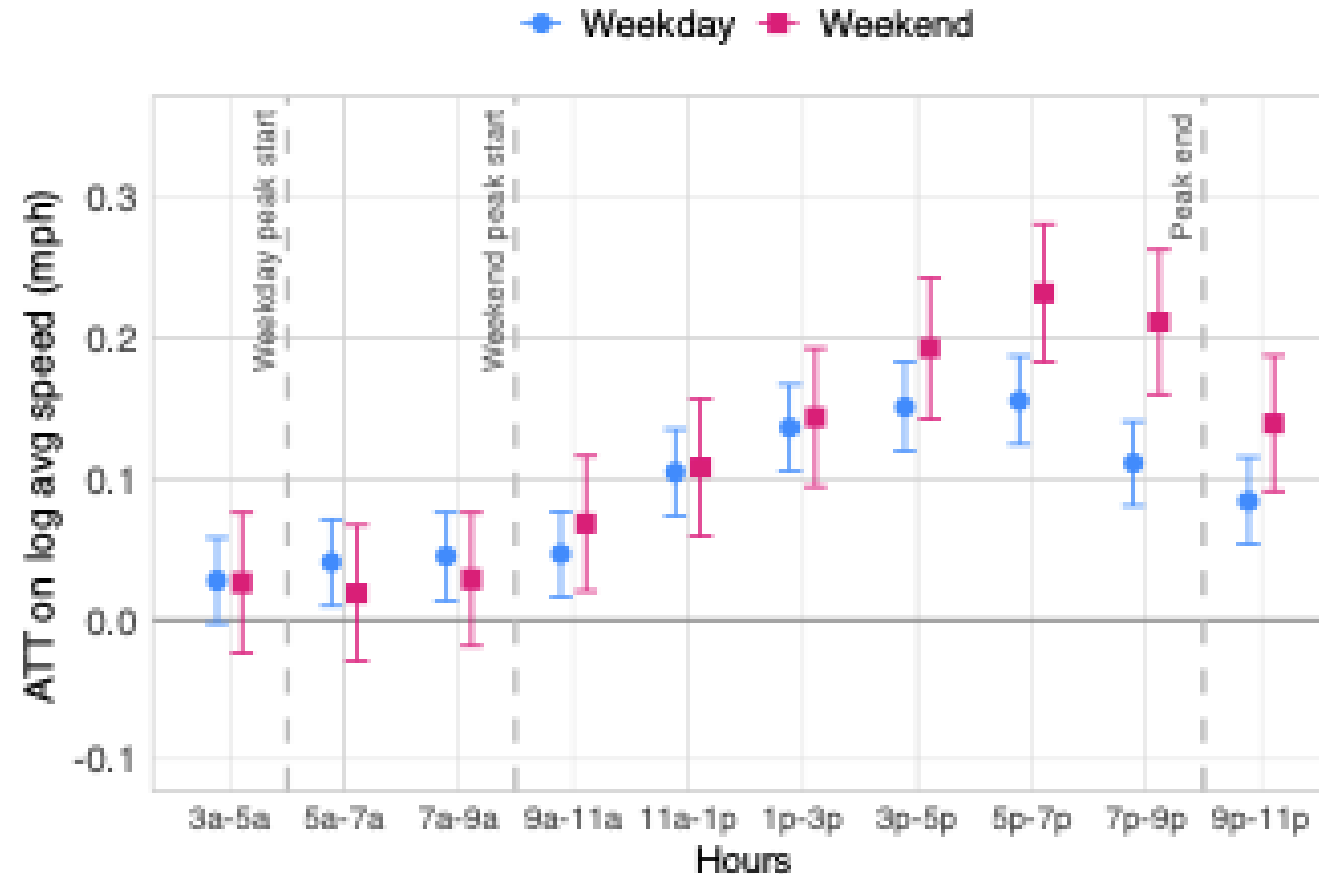
NYC congestion pricing: one year later

(c) Average segment speeds by hour of day



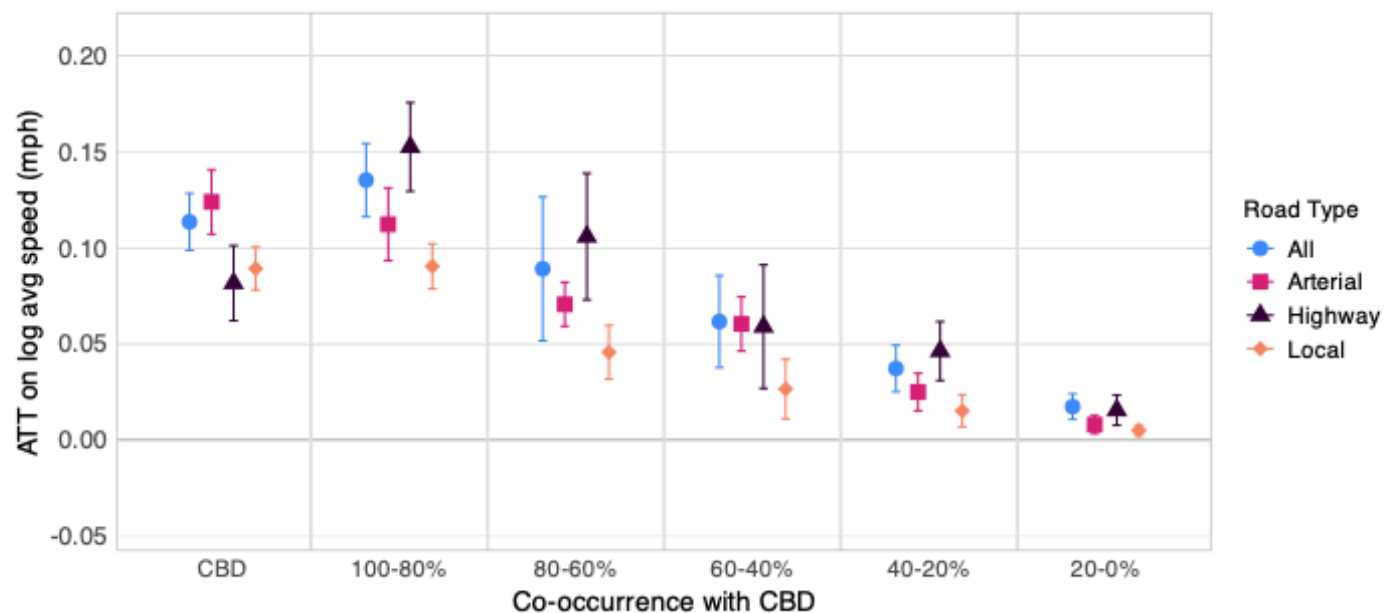
NYC congestion pricing: one year later

(d) ATT on log speeds: hour of day



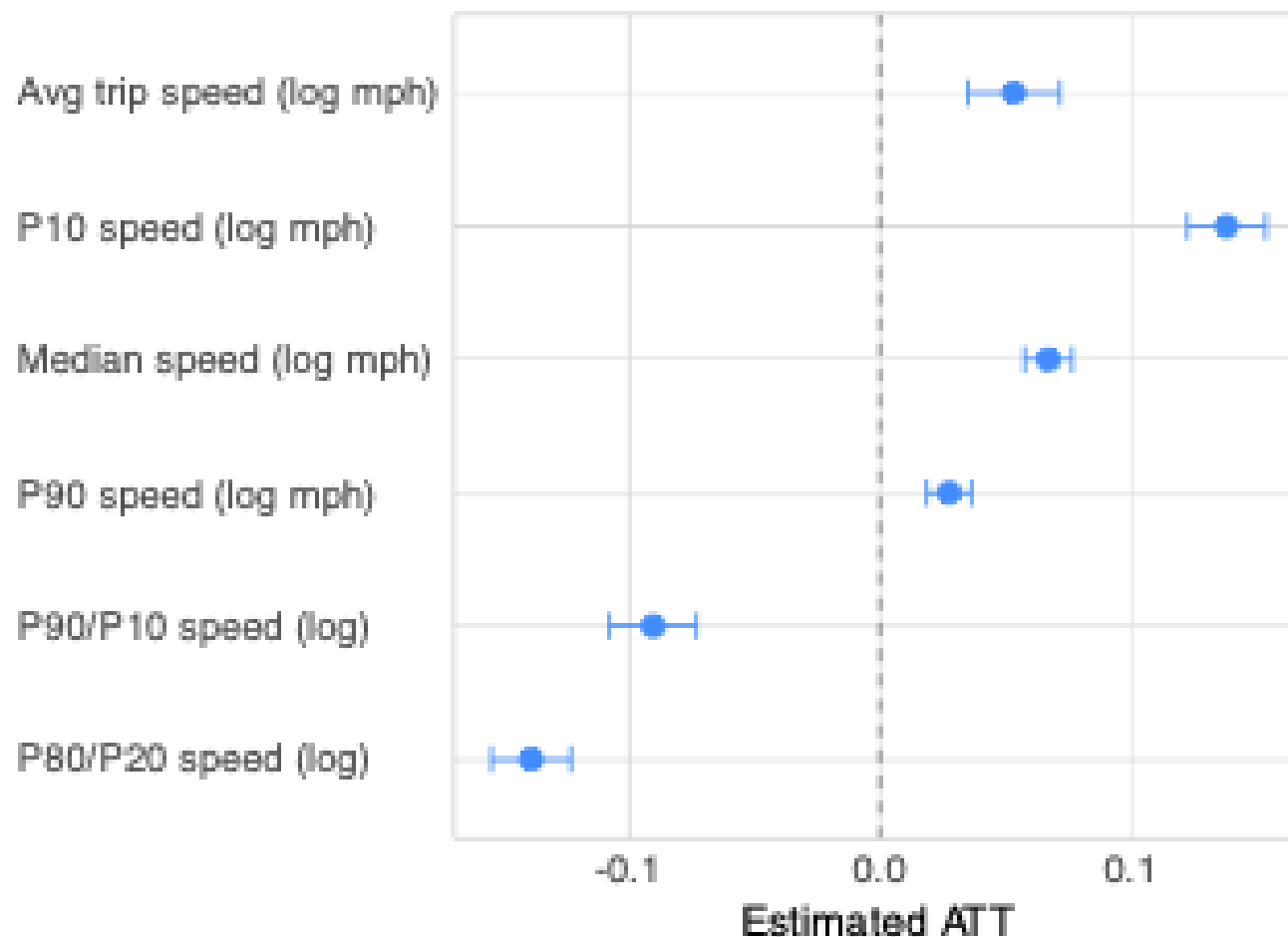
NYC congestion pricing: one year later

Figure 4: Effect on Speeds by Co-Occurrence and Road Type



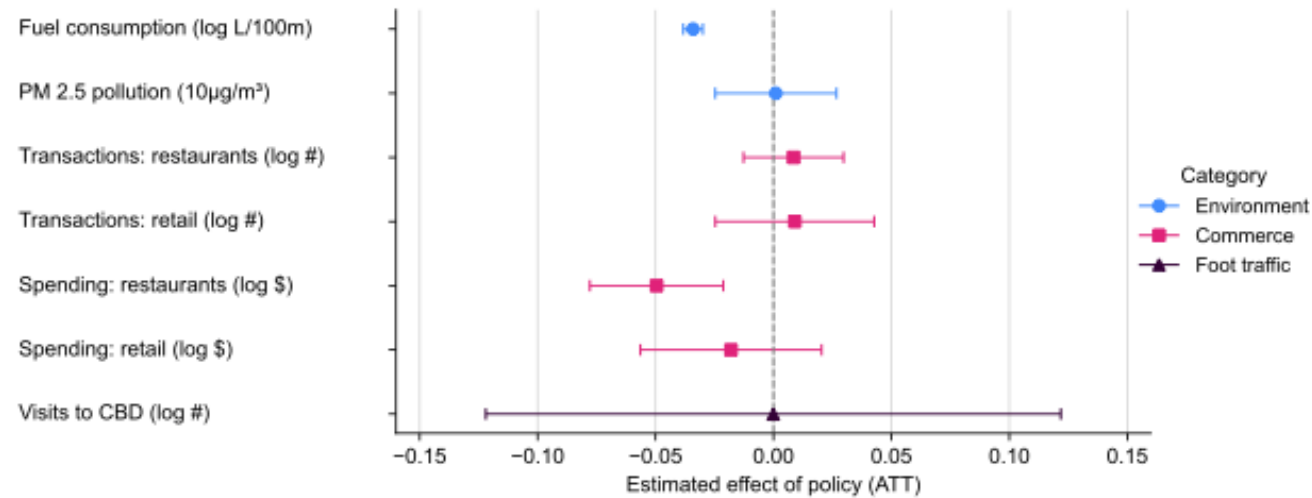
NYC congestion pricing: one year later

(b) ATT on other traffic outcomes



NYC congestion pricing: one year later

Figure 3: Effects on Other Outcomes in the CBD



Taxes don't always go smoothly

Pigouvian taxes are economically efficient, but can face **political backlash**

In 2018, France proposed a fuel tax increase to reduce carbon emissions

- 6.5 cents/liter on diesel
- 2.9 cents/liter on gasoline

The response: **Yellow Vest protests** (Gilets Jaunes)

France's Yellow Vest protests

What happened:

- 300,000+ protesters across France
- Blocked roads, occupied roundabouts
- Violent clashes in Paris
- 10 deaths, thousands injured over months of protests

Why?

- Rural and suburban households depend on cars
- Tax seen as burden on working class while wealthy unaffected
- "End of the world vs end of the month"

France's Yellow Vest protests



Lessons for environmental taxes

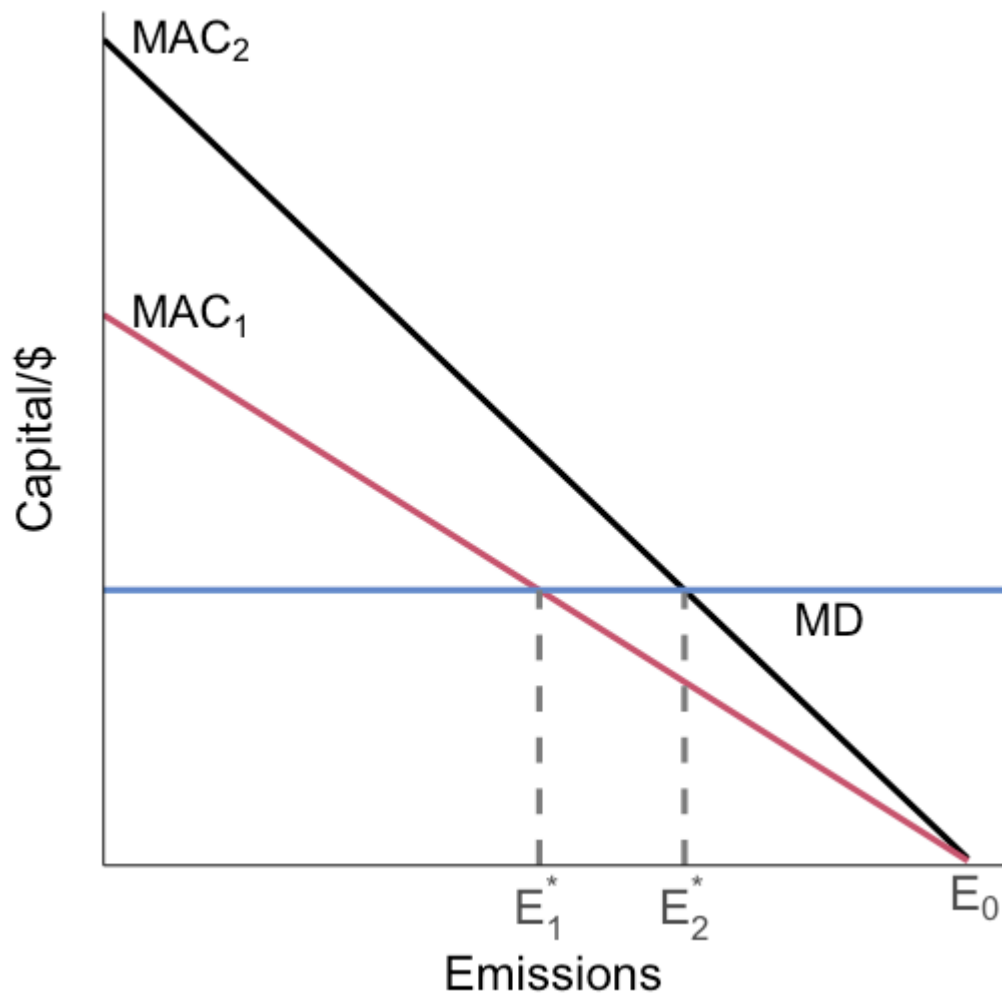
The Yellow Vests show that **distributional concerns matter**

Successful environmental taxes often include:

- Revenue recycling to affected households
- Exemptions for low-income groups
- Gradual phase-in periods
- Clear communication of benefits

NYC's congestion pricing includes income-based exemptions for this reason

Optimal policy with multiple firms



Firm #2 is 'dirty': has higher MAC

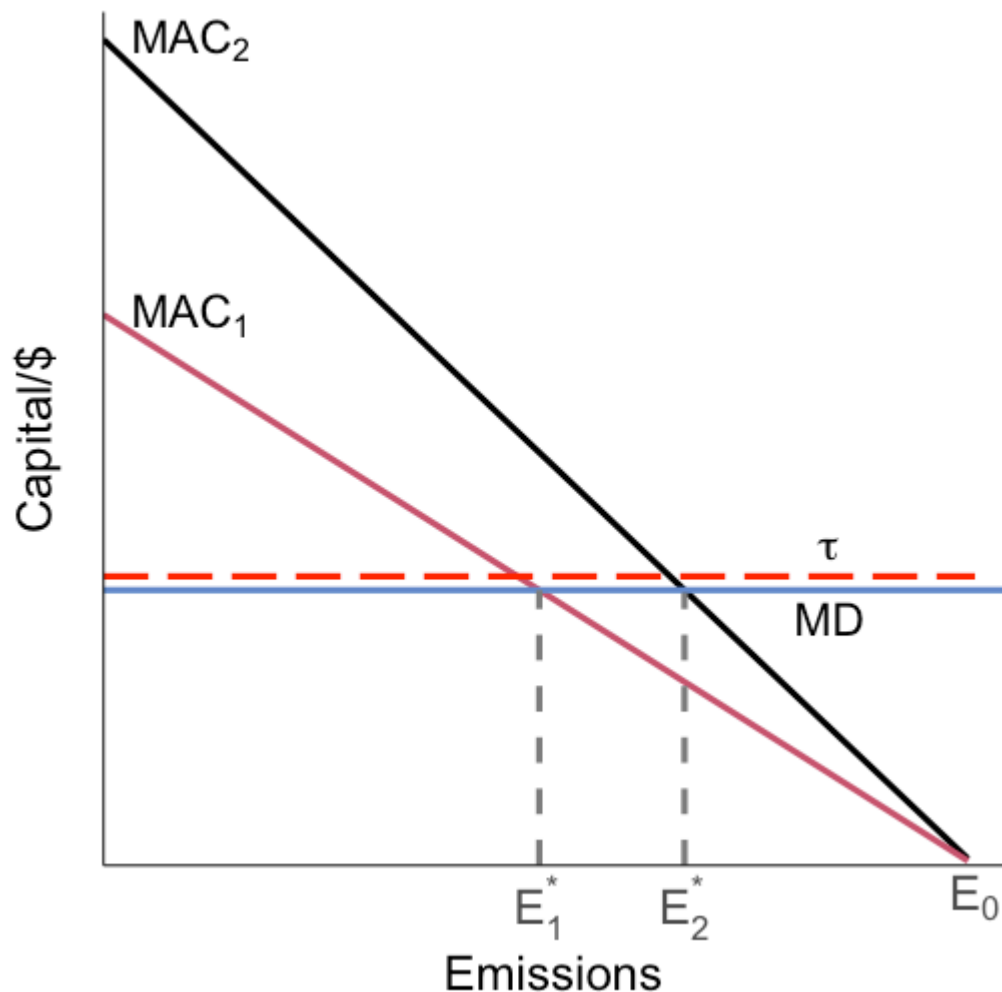
Firm #1 is 'clean': has lower MAC

If we use a regular emission standard: it has to be firm-specific!

Mandate E_1^* for 1 and E_2^* for 2

This requires **a lot** of info and political capital on behalf of the regulator

Optimal policy with multiple firms



Regulating multiple heterogeneous firms with a tax can be easy:

If MD is constant, then since firms select $MAC = \tau$, as long as we set $\tau = MD$, we can achieve the efficient outcome ($MAC = MD$) without knowing anything about the firms!

Optimal policy with multiple firms

Taxes also achieve the **cost-effective** outcome: achieving a given emission level at least-cost

Let's see why

Suppose we want to minimize the total cost of achieving emission level $\bar{E}$ by abating across two different sources, plant 1 and plant 2

The plants have abatement cost functions: $C_1(E_1)$ and $C_2(E_2)$

Write down the regulator's problem

Optimal policy with multiple firms

$$\min_{E_1, E_2} C_1(E_1) + C_2(E_2) \quad \text{subject to: } E_1 + E_2 = \bar{E}$$

Solve the constraint for $E_2 = \bar{E} - E_1$ so we have a simpler problem:

$$\min_{E_1} C_1(E_1) + C_2(\bar{E} - E_1)$$

Take the first-order condition to find what is necessary for a cost minimum:

$$C'_1(E_1) + C'_2(\bar{E} - E_1) \times (-1) = 0$$

Optimal policy with multiple firms

This gives us:

$$\underbrace{-C'_1(E_1)}_{\text{MAC}_1} = \underbrace{-C'_2(\overbrace{\bar{E} - E_1}^{E_2})}_{\text{MAC}_2}$$

The marginal abatement costs across the sources must be equal at the cost-effective pollution level

This is called the **equimarginal principle**

Optimal policy with multiple firms

Taxes **always** achieve the equimarginal principle and get us the given amount of emission reductions at least-cost

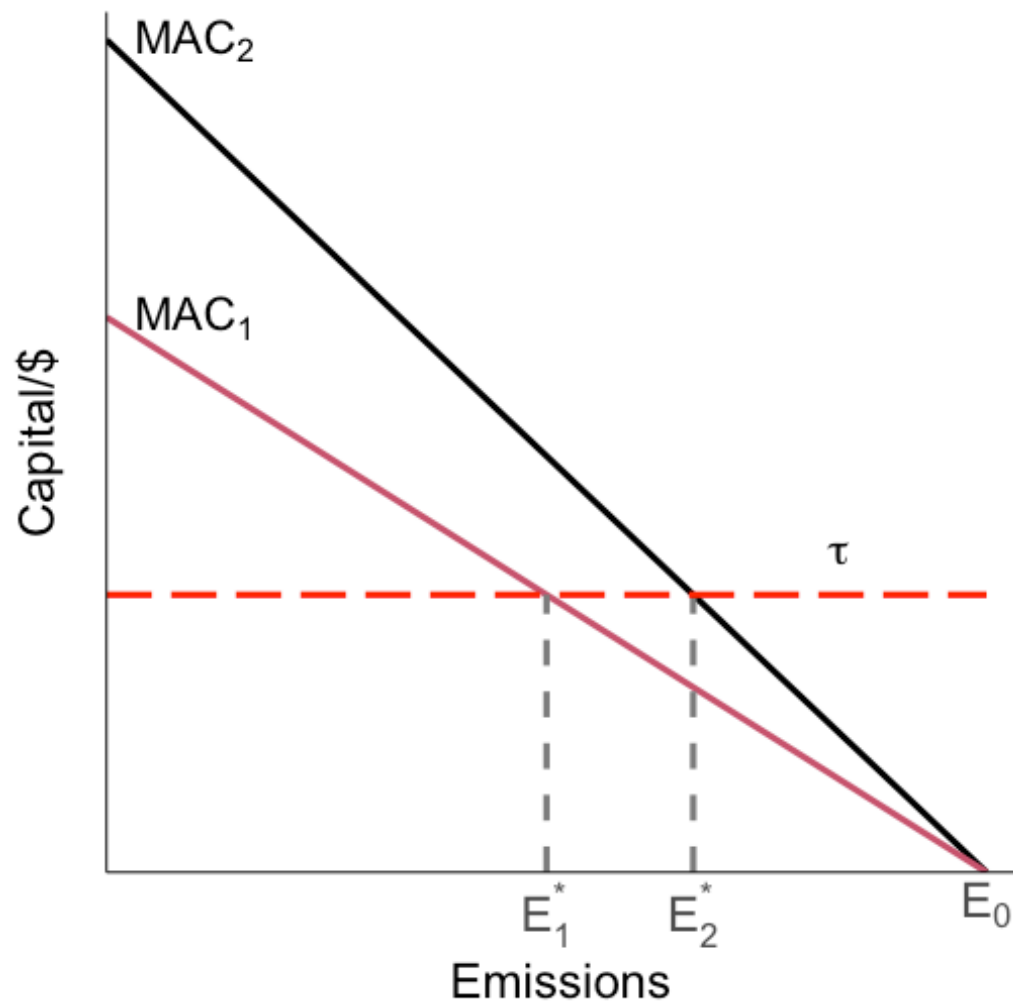
Why?

We know firms optimally select MAC equal to the emission tax

This means all firms' MACs are equal!

Even if we don't set the tax equal to MD, whatever emission reduction we get will be as cheap as possible

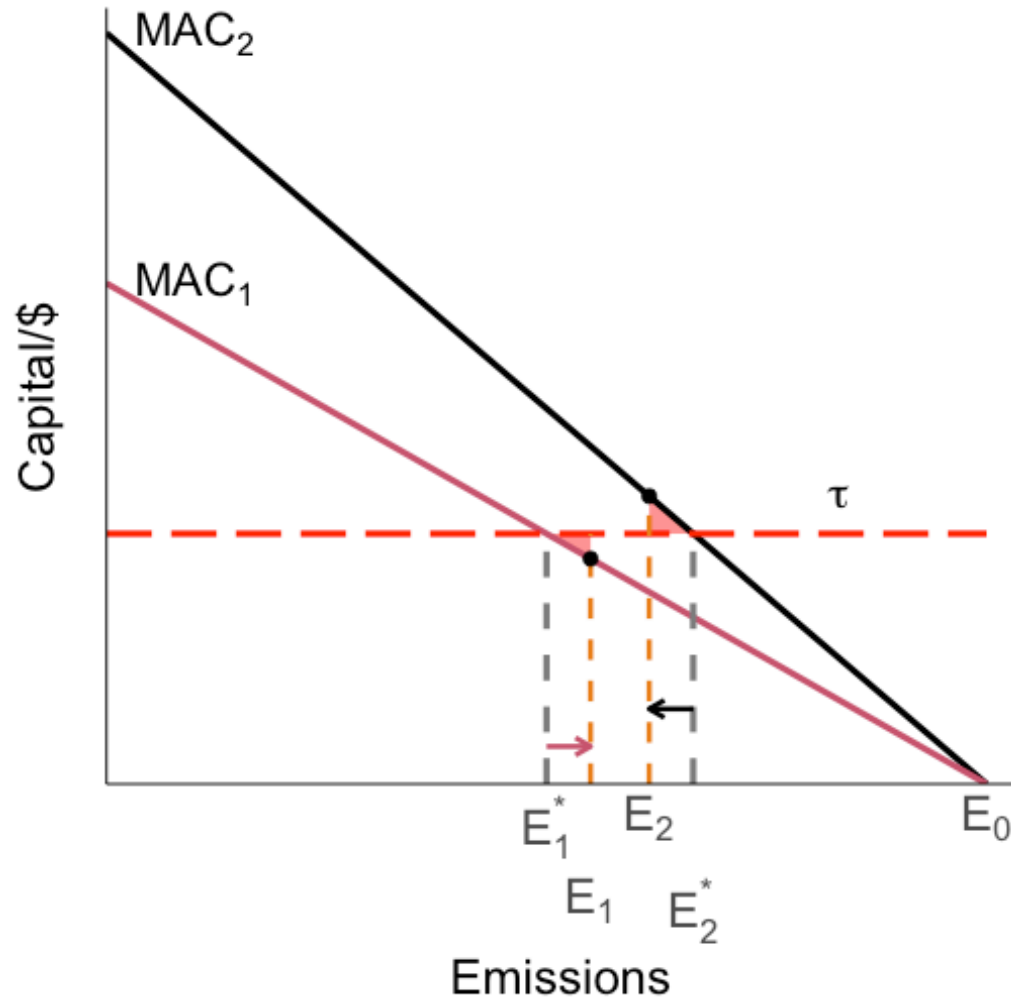
Optimal policy with multiple firms



If standards are applied uniformly across heterogeneous firms, costs are not minimized

If firms face a common emissions price, abatement is reallocated toward low-MAC sources

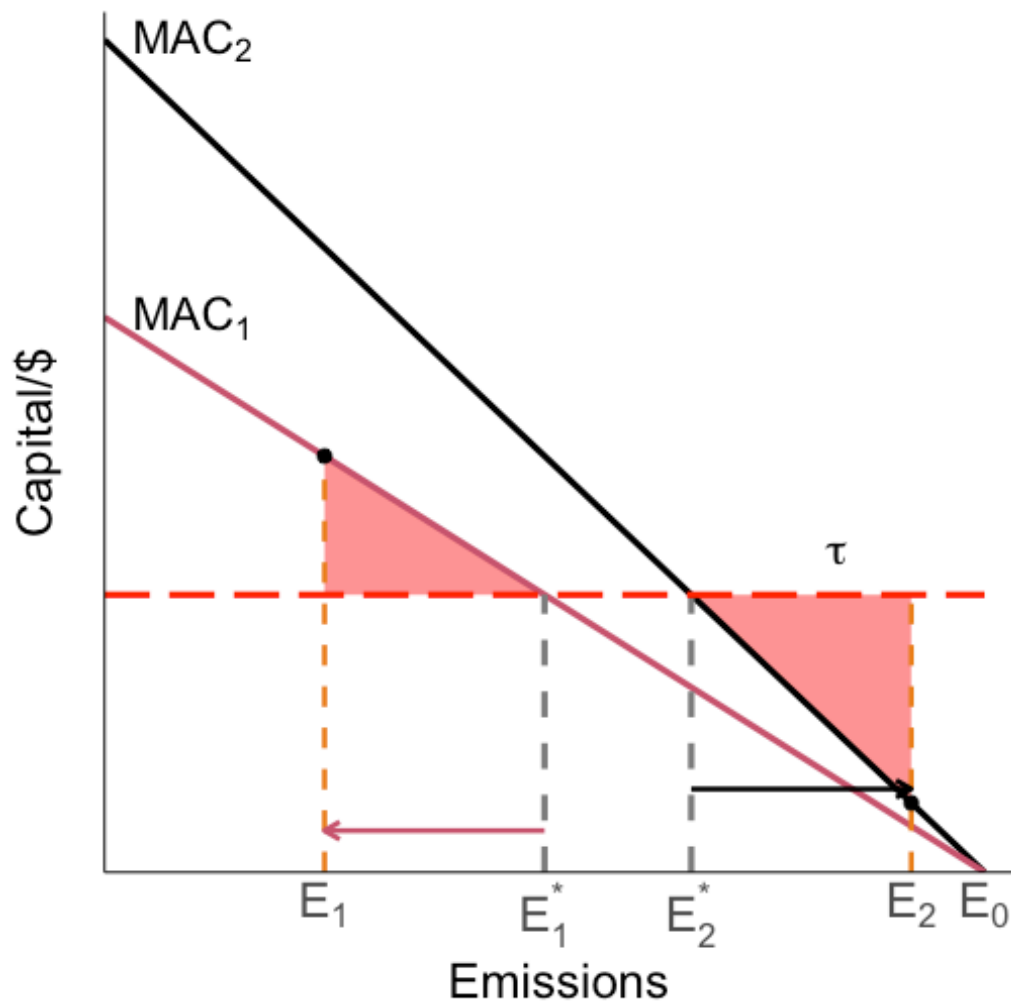
Deviating from the equimarginal principle



The **red area** is DWL relative to the cost-minimizing allocation

Shift emissions back toward firm 2 to equalize MACs

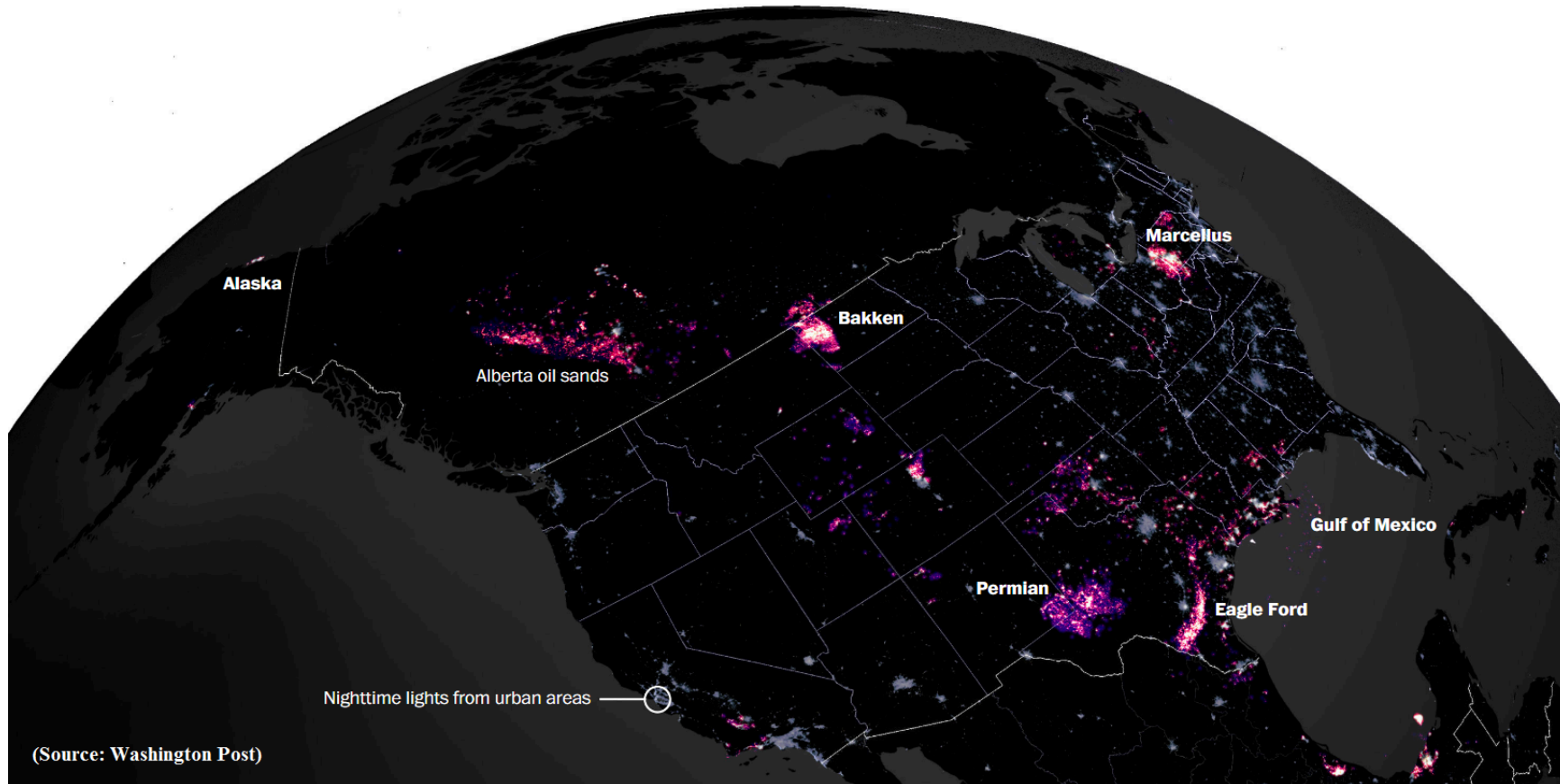
Deviating in the other direction



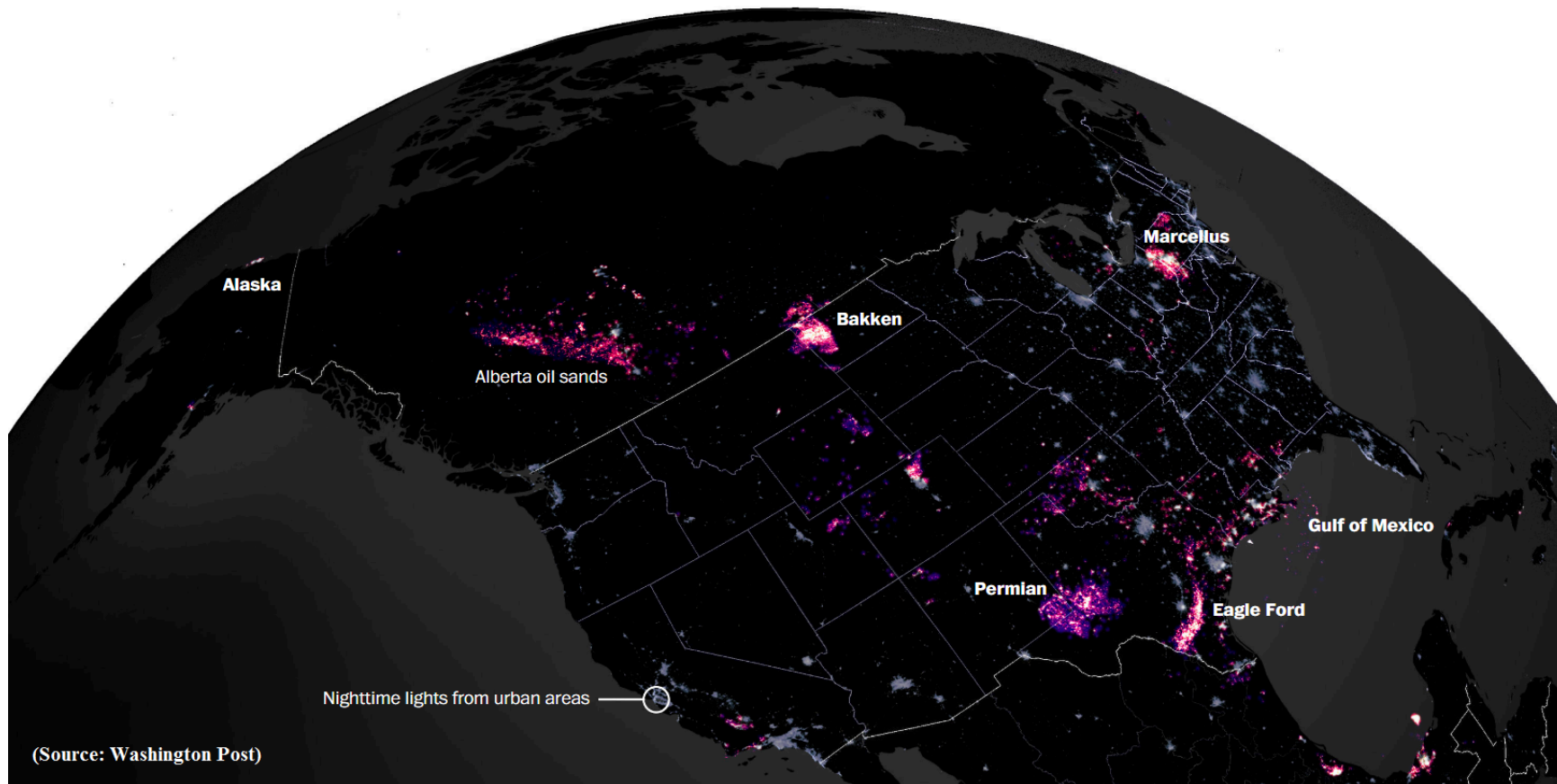
The **red area** is DWL relative to the cost-minimizing allocation

Shift emissions back toward firm 1 to equalize MACs

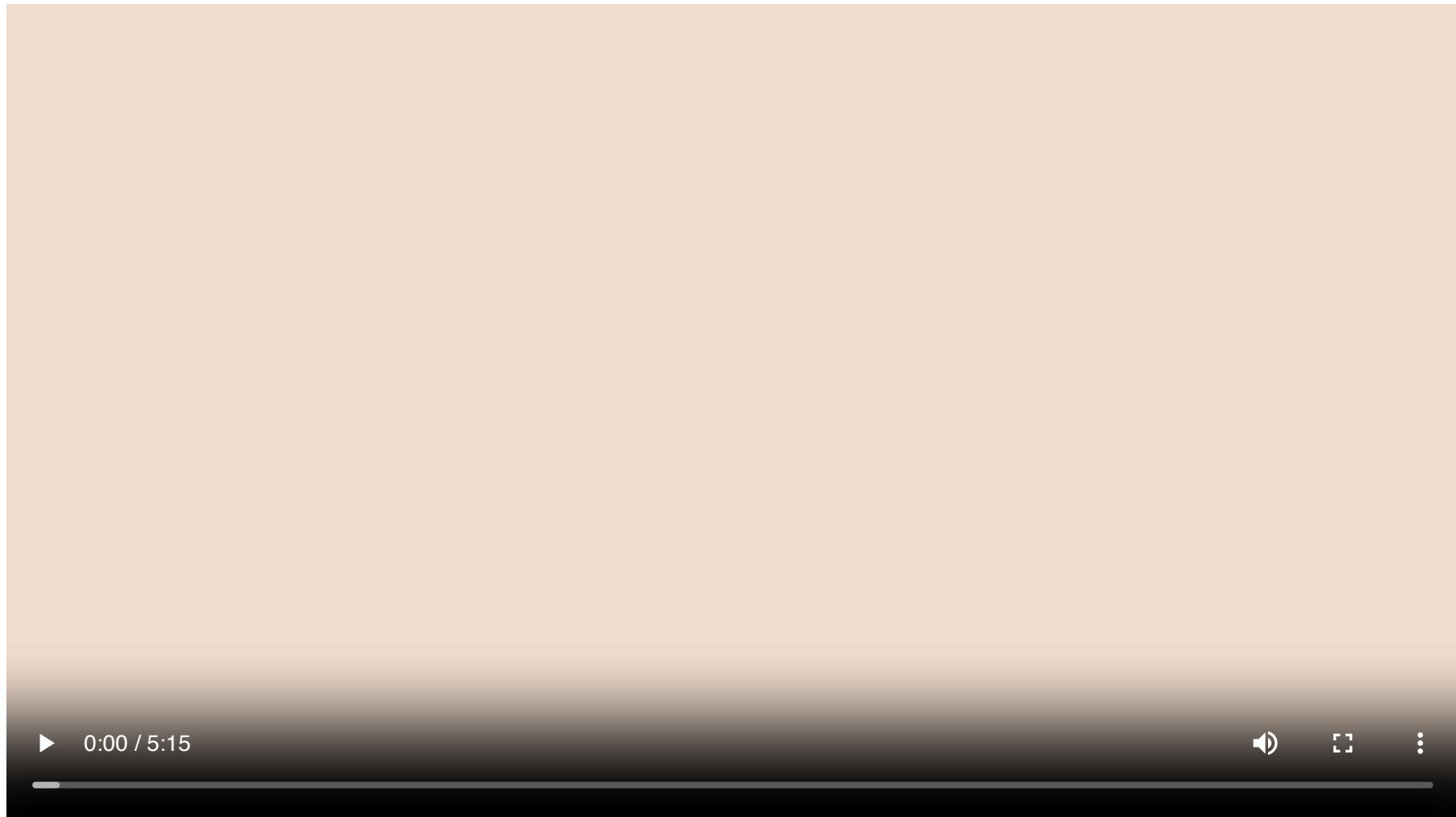
Taxes vs Standards



Taxes vs Standards



Emission taxes



Unconventional oil



(Andrew Burton/Getty Images)

Unconventional oil



(Andrew Burton/Getty Images)

Unconventional oil



(Andrew Burton/Getty Images)

Flaring regulations in North Dakota

North Dakota Industrial Commission established firm-level flaring limits beginning in October 2014

North Dakota Industrial Commission Order 24665 Policy/Guidance **Version 041718**

Policy Goals:

- 1) reduce the flared volume of gas
- 2) reduce the number of wells flaring
- 3) reduce the duration of flaring from wells

The Commission establishes the following gas capture goals:

- 74% October 1, 2014 through December 31, 2014
- 77% January 1, 2015 through March 31, 2016
- 80% April 1, 2016 through October 31, 2016
- 85% November 1, 2016 through October 31, 2018
- 88% November 1, 2018 through October 31, 2020
- 91% beginning November 1, 2020

Flaring regulations in North Dakota

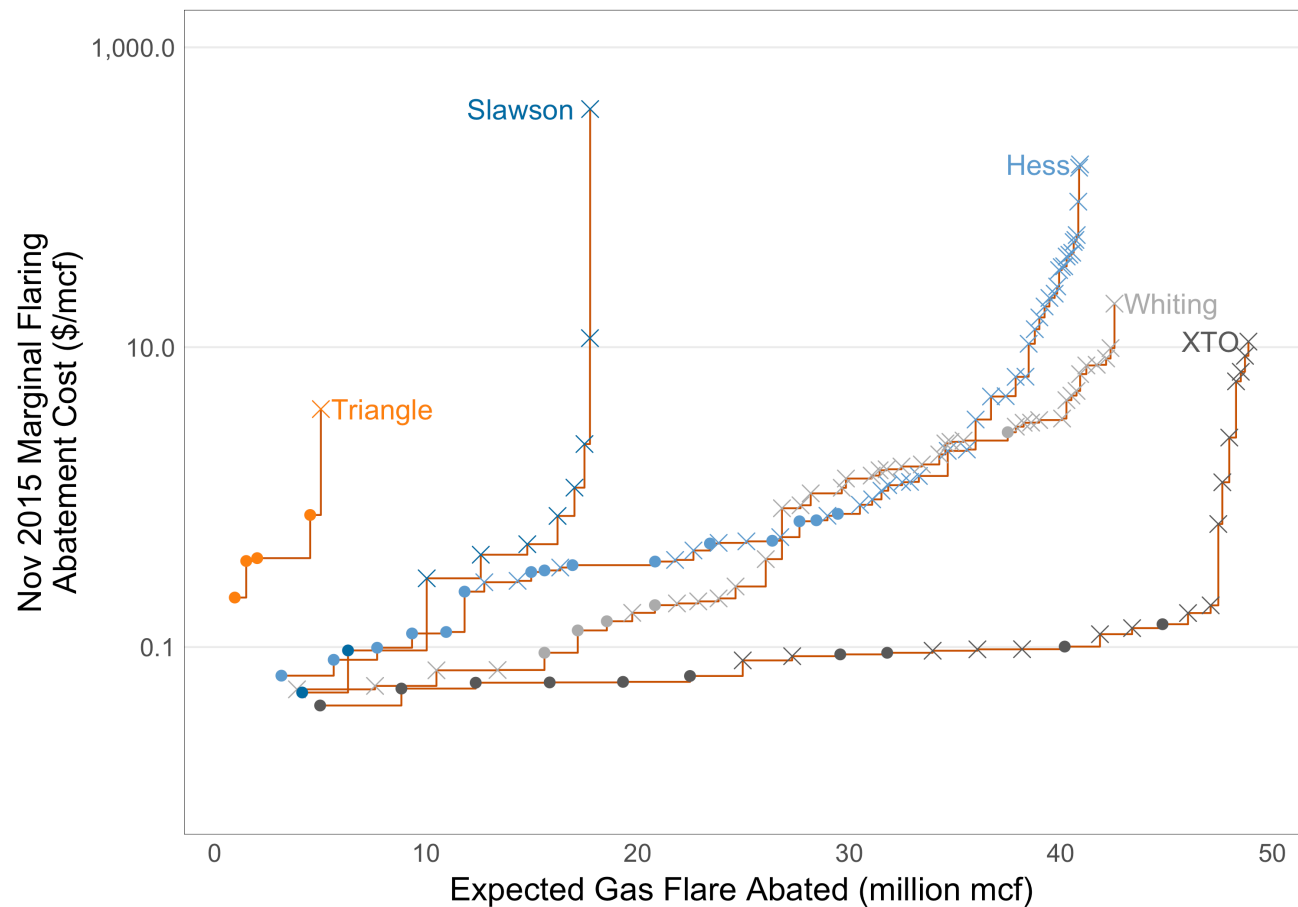
This is a **standard**: a mandate on the quantity of gas captured

It was applied uniformly across firms

So with heterogeneous MACs, this generally violates the equimarginal principle and is not cost-effective

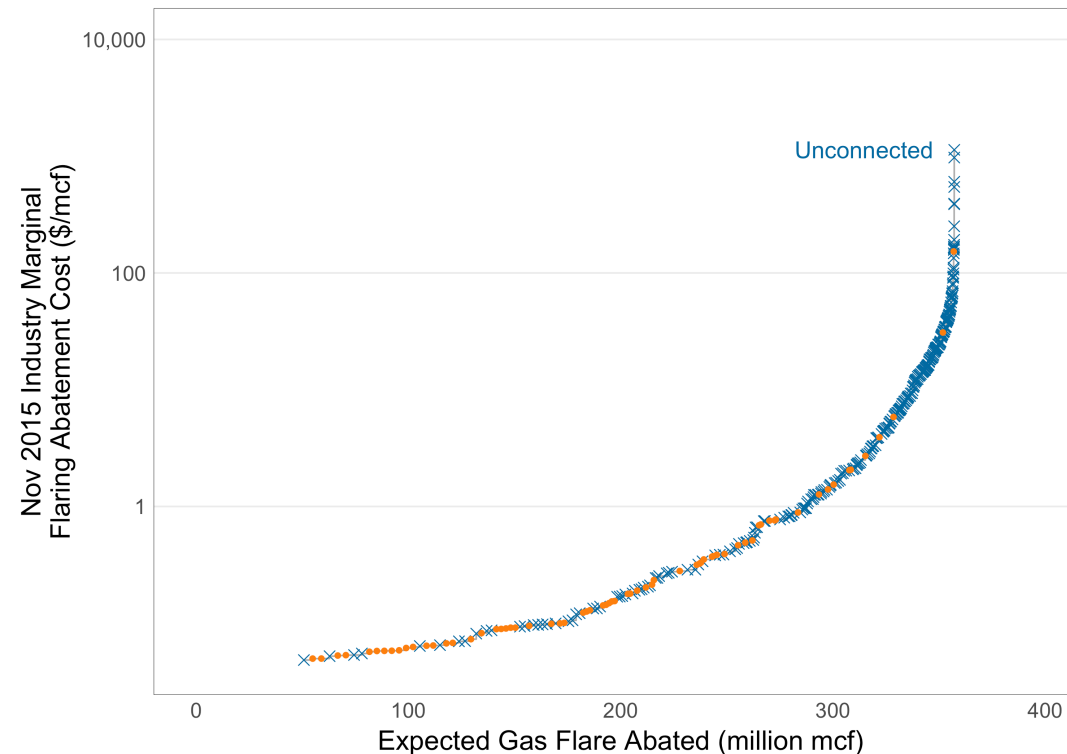
Real world MACs

Firms appear to be cost-minimizing to comply



Real world MACs

If we add up the MACs to the market-wide MAC, we aren't always abating at the cheapest wells (because the standard is uniform across firms)



What if we used a tax?

What if we set a tax instead of a standard?

Suppose we just used the existing *public* lands royalty rate

We could capture 99% of the gas as the regulation, but 50% of the cost

Almost everyone is better off

The majority of firms are better off with a tax

